

AGL Machine Operating Time Forecasting

- **Introduction:**

Africa Gulf Logistics (AGL) operates two major port terminals: Abidjan Terminal Logistics (ATL) and Côte d'Ivoire Terminal (CIT). These ports are crucial hubs for regional and international trade, handling a variety of vessels daily. However, AGL has various port terminal KPIs, one of them is Ship to Ship cranes (STS) operating time. Operating time refers to the total time a machine has been active and operating on ships. With increasing traffic and the diverse nature of cargo, optimizing operating time has become a complex logistical issue. AGL wants to forecast the volume of change in STS cranes operating time through days, weeks, months, and years.

- **Dataset:**

The dataset is a csv file with 14 columns as Eqp_Type_Custom is the name of each machine, the 14 columns names are:

0 Eqp_Key	1 Call_Key
2 Operating_Time	3 Networking_Time
4 First_Move_Eqp_Dt_Year	5 First_Move_Eqp_Dt_YearMonth
6 First_Move_Eqp_Dt_YearMonthDay	7 Last_Move_Eqp_Dt_Year
8 Last_Move_Eqp_Dt_YearMonth	9 Last_Move_Eqp_Dt_YearMonthDay
10 Last_Move_Eqp_Dt_WeekThursday	11 Eqp_Call_Key
12 Eqp_ID	13 Eqp_Type_Custom

Here's a breakdown of columns names:

Eqp: Equipment

CALL: Each arrival or departure of a vessel is recorded as a Call

Terminal 6
☐ P1 ☐ P2 ☒ P3 ☐ P4 ☐ P5 ☐ P6 ☒ P7 ☐ P8

Terminal 28
☐ CR1 ☒ CR2 ☐ CR3 ☐ CR4 ☐ CR5 ☐ CR6

P.S. Those are machine names of each Port Terminal, 6 is ATL, 28 is CIT.

- Requirements:

You'll develop a deployed model to forecast the operating time of selected machines by days and weeks, consisting of an API that takes selected machine IDs plus the type of forecasting either by days or weeks, then returns an image of the forecasting graph of their operating time.